

ULTRON 2.7 Phase 7 Execution Report

Cyberpunk visual interface and local brain API

Project	ULTRON 2.7
Phase	Phase 7: Visual Interface
Workspace	Local workspace checkout: U2.7
Status	Implemented and verified

Executive Summary

Phase 7 gives ULTRON 2.7 a local web interface instead of a CLI-only experience. The interface presents a cyberpunk black-and-green scene with a Three.js plasma sphere, subtle particles, live subtitles, typed command input, and explicit visual states for listening, thinking, speaking, and idle. The UI connects to ULTRON's existing brain/runtime through a local API. It does not bypass schema validation, policy, execution restrictions, or audit logging.

Phase 7 Scope

- Create a real local web interface backed by the ULTRON runtime.
- Build a responsive Three.js sphere scene with green plasma lines, eclipse rings, particles, glow, and state transitions.
- Add subtitle rendering and a subtitle on/off toggle.
- Add typed command input with Send button and Enter-key behavior.
- Add local API endpoints for commands, status, memory, subtitles, and development visual state control.
- Update tests, README, architecture notes, and implementation documentation.

What Was Implemented

Area	File(s)	Result
Local API	src/ultron27/web_server.py	Serves the static UI and exposes command, status, memory, subtitle, and state endpoints.
Launcher	scripts/run_phase7_ui.py, pyproject.toml	Adds a script launcher and the ultron-ui package entry point.
Frontend shell	web/index.html, web/styles.css	Adds the full-screen black-and-green interface, subtitle panel, controls, and command bar.

Area	File(s)	Result
3D scene	web/app.js, web/vendor/three.module.min.js	Adds the Three.js sphere, particles, rings, plasma arcs, and state transitions.
Smoke check	scripts/phase7_visual_smoke.mjs	Captures screenshots when a local Playwright browser binary is available.
Docs	README.md, docs/architecture.md, docs/phase7_visual_interface.md	Documents UI, endpoints, safety boundary, and run instructions.
Tests	tests/test_pipeline.py	Adds Phase 7 API, subtitle, static asset, and command-processing coverage.

Visual State Design

State	Animation	User Impression
Listening	Sphere contracts slightly, glow tightens, breathing ring calms, particle motion slows.	ULTRON is focused on the user's next command.
Thinking	Green orbit rings spin around the sphere with crescent-like eclipse motion.	ULTRON is analyzing and planning the request.
Speaking	Sphere expands, plasma lines brighten, outer glow intensifies, particles push outward.	ULTRON is delivering a result.
Idle	Sphere remains alive with restrained motion and a stable field.	The system is online and ready.

Runtime Safety Boundary

Layer	Responsibility	Safety Property
Web UI	Collect typed commands and display state/subtitles.	No shell access and no direct executor calls.
Local API	Translate HTTP requests into brain calls and response snapshots.	Only exposes typed endpoints.
Brain	Plan and execute task steps through the assistant runtime.	Cannot skip planner, validator, policy, or executor.
Runtime	Validate tool calls, apply policy, execute allowed tasks, and audit.	High-risk and blocked actions remain gated.

Verification Results

Check	Command	Result
Unit tests	<code>\$env:PYTHONPATH='src'; python -m unittest discover -s tests</code>	35 tests passed.

Check	Command	Result
Compile check	<code>python -m py_compile src/ultron27/web_server.py scripts/run_phase7_ui.py</code>	Backend and launcher compiled successfully.
Dataset evaluation	<code>python scripts/evaluate_dataset.py --split test</code>	Intent 1.0, tool 1.0, argument exact match 0.9614, confirmation 1.0.
Safety regression	<code>python scripts/evaluate_safety_regression.py</code>	362 cases, tool accuracy 1.0, policy action accuracy 1.0.
Browser smoke	<code>node scripts/phase7_visual_smoke.mjs</code>	Clean fallback when the Playwright browser binary is unavailable in the sandbox.

Recommended Next Step

- Move to Phase 8 by adding voice input and voice output as adapters around the same local API and brain/runtime.
- Map voice activity to the existing listening, thinking, and speaking visual states.
- Keep confirmation prompts visible in the interface before enabling broader real-world task execution.